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Problem Statement #08

Domain Campus & Academic Operations

Digital Campus Library Reservation Gateway

Problem Statement

A campus library platform that lets members search the physical ISBN catalog, join a fair waitlist for copies that are currently on loan, get notified the moment a hold becomes collectable, and stay on top of overdue fines. The two people who touch the system are the **Student Member** who borrows and reserves, and the **Head Librarian** who keeps the catalog and fine records in order.

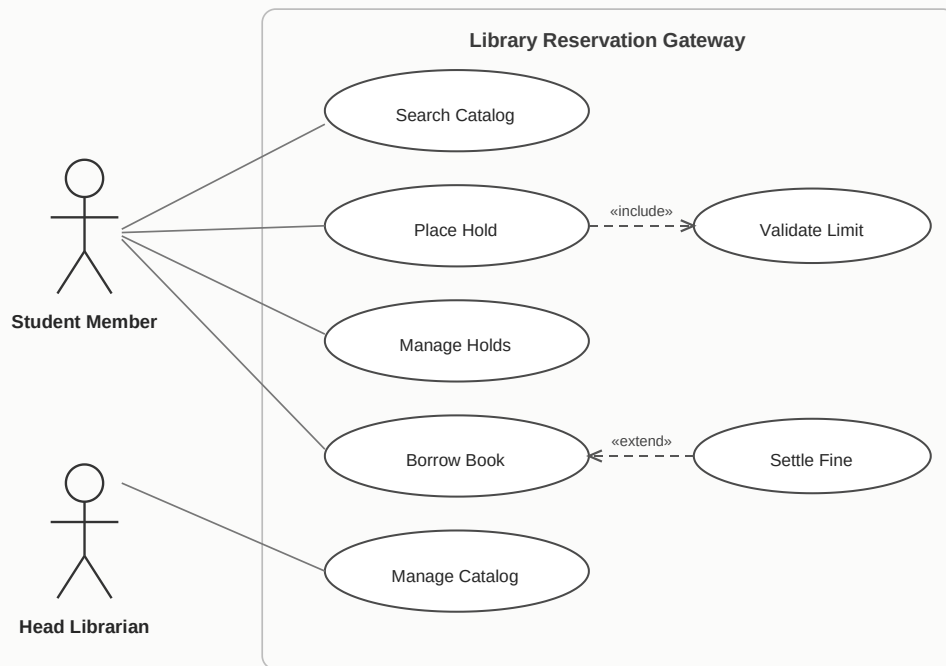
1 Requirements

ID	DESCRIPTION	PRIORITY	ACCEPTANCE CRITERIA	RATIONALE
FR-001	The system shall let a member search the ISBN repository and place a hold on a title that is currently on loan, adding them to a first-in-first-out waitlist for that title.	High	PASS: the member lands in the correct queue position for that ISBN. FAIL: a member already at their borrowing limit is allowed to reserve anyway.	Reservations are the core reason this gateway exists; a fair FIFO order is what members trust it for.
FR-002	When a reserved copy is returned and reaches the front of its queue, the system shall automatically notify the next member and hold the copy for a fixed collection window.	High	PASS: only the front-ofqueue member is notified, and the copy is flagged as reserved for them. FAIL: the copy is offered to a later member while an earlier one is still waiting.	Without an automatic hand-off the waitlist stalls and returned books sit unclaimed on the shelf.
FR-003	The system shall record every checkout and return, and when a book is returned past its due date it shall calculate and log an overdue fine against the member's account.	High	PASS: a book returned three days late produces a fine that matches the day-rate schedule. FAIL: an on-time return is charged a fine.	Fine tracking is what keeps circulation moving and gives the librarian an accurate ledger.

FR-004	The system shall allow a member to view their active holds with live queue positions and to cancel any hold before the copy is collected.	Medium	PASS: cancelling a hold removes the member and shifts everyone behind them up by one. FAIL: a cancelled hold still shows the member as waiting.	People change their minds; letting them step out cleanly keeps the queue honest for everyone else.
FR-005	The system shall let the Head Librarian add, edit, and retire ISBN catalog records, including the number of physical copies held for each title.	Medium	PASS: a newly added ISBN is immediately searchable by members. FAIL: a retired title still accepts new hold requests.	The catalog is only as useful as it is current, and only the librarian should be able to reshape it.
ID	DESCRIPTION	PRIORITY	ACCEPTANCE CRITERIA	RATIONALE
NFR-001	Type: Performance & Security. A catalog search across 50,000 titles shall return paginated results in under 150 ms, and all such queries shall run over authenticated, encrypted sessions.	High	PASS: benchmark runs stay under 150 ms at simulated peak load with TLS enforced. FAIL: latency drifts above target or an unauthenticated query returns data.	Search is the first thing every member does; if it feels slow or leaks, nothing else matters.
NFR-002	Type: Reliability & Availability. The reservation and notification services shall stay available at least 99.5% of each month, and no confirmed hold shall be lost during a service restart.	Medium	PASS: holds placed before a forced restart are all present and correctly ordered afterwards. FAIL: a restart drops a confirmed hold or scrambles queue order.	A waitlist that forgets people the moment the server hiccups would quietly erode all trust in the system.

2 UML Use-Case Diagram

Both actors, the primary use cases, and the two required relationships — one «include» and one «extend».



Association — solid line between actor and use case **«include»** — Place Hold always validates the borrowing limit

«extend» — Settle Fine extends Borrow Book, stepping in only when a fine is owed

3 Use-Case Flow — Place Hold on a Loaned Book

Core use case UC-02, covering the happy path and one realistic detour.

USE CASE	UC-02 · Place Hold on a Loaned Book — Primary actor: Student Member
PRECONDITIONS	The member is signed in, their account is in good standing, and the title they want is currently on loan to someone else.
POSTCONDITIONS	The member holds a confirmed place in the title's FIFO waitlist, and the system knows to notify them once a copy frees up.
MAIN SUCCESS SCENARIO	1. The member searches the catalog by title, author, or ISBN. 2. The system shows matching titles and marks this one as <i>on loan</i>. 3. The member selects the title and chooses <i>Place Hold</i>. 4. The system checks that the member is below their borrowing limit («include» <i>Validate Limit</i>). 5. The system adds the member to the back of that title's waitlist. 6. The system confirms the hold and shows the member their queue position.

ALTERNATE FLOW

4A — BORROWING LIMIT
REACHED

1. At step 4 the system finds the member is already at their maximum number of active loans and holds.
2. The system declines the request and explains that the limit has been reached.
3. It suggests returning or cancelling an existing item, then returns the member to the search results. No hold is created.